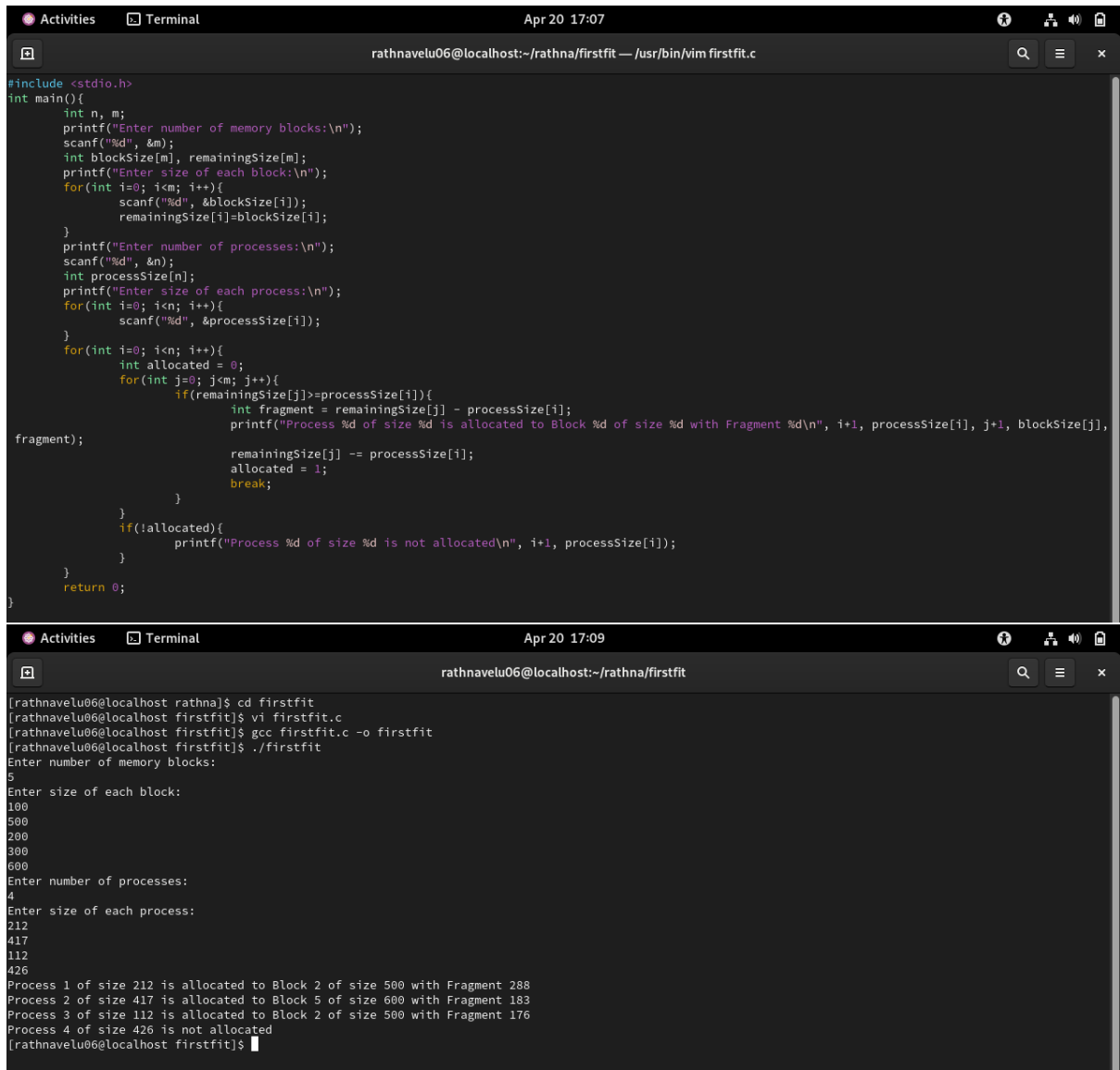


OPERATING SYSTEMS CONCEPT – LAB

EXP – 09: CONTIGUOUS MEMORY ALLOCATION

FIRST FIT



```
#include <stdio.h>
int main(){
    int n, m;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);
    int blockSize[m], remainingSize[m];
    printf("Enter size of each block:\n");
    for(int i=0; i<m; i++){
        scanf("%d", &blockSize[i]);
        remainingSize[i]=blockSize[i];
    }
    printf("Enter number of processes:\n");
    scanf("%d", &n);
    int processSize[n];
    printf("Enter size of each process:\n");
    for(int i=0; i<n; i++){
        scanf("%d", &processSize[i]);
    }
    for(int i=0; i<n; i++){
        int allocated = 0;
        for(int j=0; j<m; j++){
            if(remainingSize[j]>=processSize[i]){
                int fragment = remainingSize[j] - processSize[i];
                printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n", i+1, processSize[i], j+1, blockSize[j],
fragment);
                remainingSize[j] -= processSize[i];
                allocated = 1;
                break;
            }
        }
        if(!allocated){
            printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
        }
    }
    return 0;
}
```

```
[rathnavelu06@localhost rathna]$ cd firstfit
[rathnavelu06@localhost firstfit]$ vi firstfit.c
[rathnavelu06@localhost firstfit]$ gcc firstfit.c -o firstfit
[rathnavelu06@localhost firstfit]$ ./firstfit
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 2 of size 500 with Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500 with Fragment 176
Process 4 of size 426 is not allocated
[rathnavelu06@localhost firstfit]$
```

BEST FIT

```
Activities Terminal Apr 20 17:04
rathnavelu06@localhost:~/rathna/bestfit — /usr/bin/vim bestfit.c

#include <stdio.h>
int main(){
    int m, n;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);
    int blockSize[m], remaining[m];
    printf("Enter size of each block:\n");
    for(int i=0; i<m; i++){
        scanf("%d", &blockSize[i]);
        remaining[i]=blockSize[i];
    }
    printf("Enter number of processes:\n");
    scanf("%d", &n);
    int processSize[n];
    printf("Enter size of each process:\n");
    for(int i=0; i<n; i++){
        scanf("%d", &processSize[i]);
    }
    for(int i=0; i<n; i++){
        int bestIndex=-1;
        for(int j=0; j<m; j++){
            if(remaining[j]>=processSize[i]){
                if(bestIndex==-1 || remaining[j]<remaining[bestIndex]){
                    bestIndex=j;
                }
            }
        }
        if(bestIndex!=-1){
            int fragment=remaining[bestIndex]-processSize[i];
            printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n", i+1, processSize[i], bestIndex+1, blockSize[bestIndex], fragment);
            remaining[bestIndex]-=processSize[i];
        } else{
            printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
        }
    }
    return 0;
}
```

```
Activities Terminal Apr 20 17:06
rathnavelu06@localhost:~/rathna/bestfit

[rathnavelu06@localhost bestfit]$ vi bestfit.c
[rathnavelu06@localhost bestfit]$ gcc bestfit.c -o bestfit
[rathnavelu06@localhost bestfit]$ ./bestfit
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 4 of size 300 with Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with Fragment 174
[rathnavelu06@localhost bestfit]$
```